

INDIAN STATISTICAL INSTITUTE

Mid-Semester Examination (2025–26)

Programme: M.Math. 2nd Year

Course: Algebraic Geometry

Date: February 25, 2026

Time: 2 hours

Maximum Marks: 30

Instructions.

- This is **not** an open-notes examination; calculators are not allowed.
- Justify all steps in your arguments. Do not assume results not covered in lectures without providing a clear proof.
- Clarity and logical presentation are essential. Vague answers may lose marks. Partial solutions may receive some credit.

Notations & Conventions.

- Throughout, k denotes an algebraically closed field unless otherwise specified.

- (1) Show that each of the following sets is not algebraic: [3 marks]
(a) $\{(\cos t, \sin t, t) \in \mathbb{A}_{\mathbb{R}}^3 : t \in \mathbb{R}\}$. [3 marks]
(b) $\{(z, w) \in \mathbb{A}_{\mathbb{C}}^2 : |z|^2 + |w|^2 = 1\}$, where $|x + iy|^2 = x^2 + y^2$ for $x, y \in \mathbb{R}$.
- (2) (a) Let F be a nonconstant polynomial in $k[X_1, \dots, X_n]$. Show that $\mathbb{A}_k^n \setminus \mathcal{Z}(F)$ is infinite for $n \geq 1$, and that $\mathcal{Z}(F)$ is infinite for $n \geq 2$. [4 marks]
(b) Deduce that if $Z \subsetneq \mathbb{A}_k^n$ is a proper affine algebraic set, then $\mathbb{A}_k^n \setminus Z$ is infinite for all $n \geq 1$. [2 marks]
- (3) Show that any conic in $\mathbb{P}_k^2$ is isomorphic to $\mathbb{P}_k^1$. [6 marks]
- (4) Let C be an affine curve in $\mathbb{A}_k^3$. Prove that $\mathbb{A}_k^3 \setminus C$ is never an affine variety. [6 marks]
- (5) Let P_1, P_2, P_3 (respectively Q_1, Q_2, Q_3) be three non-collinear points in $\mathbb{P}_k^2$. Show that there exists a projective change of coordinates $T : \mathbb{P}_k^2 \rightarrow \mathbb{P}_k^2$ such that $T(P_i) = Q_i$ for $i = 1, 2, 3$. [6 marks]
- (6) Prove that $\text{Aut}_k(\mathbb{A}_k^1) = \{t \mapsto at + b \mid a \in k^\times, b \in k\}$. [6 marks]
- (7) Give an explicit example of two affine curves C and D over k and a morphism $\varphi : C \rightarrow D$ such that the induced map on Zariski topological spaces is a homeomorphism, but φ is not an isomorphism. [6 marks]
- (8) Give an example of a morphism $\varphi : \mathbb{A}_k^2 \rightarrow \mathbb{A}_k^2$ such that $\text{Im}(\varphi)$ is dense but not open. Are all fibers equidimensional? Justify your answer. [6 marks]

*** All the best ***